

EE3621: Digital Signal Processing

Lecture 25: Design of IIR filters - Bilinear transformation, matched z-transform & simple design example (III B.Tech EEE)

Comprehensive Lecture Notes & Faculty Textbook Reference

Lecture Plan & Objectives (40 Minutes Breakdown)

- **00:00 – 06:00 (6 mins):** Motivation for Bilinear Transformation: Overcoming the severe aliasing limitation of the Impulse Invariance method. Derivation of the Bilinear Transformation via trapezoidal numerical integration of continuous differential equations.
 - **06:00 – 15:00 (9 mins):** Conformal Mapping Properties: The algebraic transformation $s = \frac{2}{T} \left(\frac{1-z^{-1}}{1+z^{-1}} \right)$. Proof of unconditional stability preservation: mapping the entire left-half s-plane ($\sigma < 0$) into the interior of the unit circle ($|z| < 1$). Mapping the entire imaginary axis $j\Omega \in (-\infty, \infty)$ one-to-one onto the unit circle $\omega \in (-\pi, \pi)$ with **zero aliasing**.
 - **15:00 – 22:00 (7 mins):** Frequency Warping & The Pre-Warping Principle: Derivation of the non-linear tangent compression $\Omega = \frac{2}{T} \tan(\omega/2)$. Why digital frequencies must be pre-warped into analog prototype frequencies before filter synthesis.
 - **22:00 – 28:00 (6 mins):** The Matched Z-Transform (MZT): Heuristic mapping of poles and zeros: $(s - s_k) \rightarrow (1 - e^{s_k T} z^{-1})$. Adding zeros at $z = -1$ to match infinite frequency attenuation. Comparative evaluation of Impulse Invariance, Bilinear Transformation, and Matched Z-Transform.
 - **28:00 – 38:00 (10 mins):** Step-by-Step Worked Numerical Examples:
 - **Example 25.1:** BLT transformation of a 1st-order analog lowpass filter.
 - **Example 25.2:** Complete design of a 2nd-order digital Butterworth lowpass filter with pre-warping.
 - **Example 25.3:** Application of the Matched Z-Transform to a 2nd-order analog prototype.
 - **38:00 – 40:00 (2 mins):** Faculty Insights, Common Misconceptions, and University Semester Exam Traps.
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1 1. LEARNING OBJECTIVES

By the end of this lecture, students will be able to:

1. **Derive** the Bilinear Transformation (BLT) operator from the trapezoidal numerical approximation of integration.

2. **Prove** that BLT maps the stable left-half s -plane strictly into the interior of the unit circle in the z -plane, guaranteeing filter stability.
 3. **Explain** why the Bilinear Transformation completely eliminates the aliasing distortion inherent to Impulse Invariance.
 4. **Analyze** the origin and physical consequences of non-linear frequency warping: $\Omega = \frac{2}{T} \tan(\omega/2)$.
 5. **Perform** frequency pre-warping on digital specifications (ω_p, ω_s) to establish precise analog prototype cutoff frequencies.
 6. **Apply** the Matched Z-Transform (MZT) rule to map analog poles and zeros into discrete polynomials.
 7. **Contrast** the operational trade-offs between Impulse Invariance, Bilinear Transformation, and Matched Z-Transform.
 8. **Synthesize** digital Butterworth lowpass filters meeting strict attenuation specifications through step-by-step calculations.
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2. DERIVATION OF THE BILINEAR TRANSFORMATION

2.1 Trapezoidal Numerical Integration

Consider a simple continuous-time first-order differential system:

$$\frac{dy(t)}{dt} = x(t) \iff sY(s) = X(s) \implies Y(s) = \frac{1}{s}X(s) \quad (1)$$

Integrating both sides from $t = (n-1)T$ to $t = nT$:

$$y(nT) - y((n-1)T) = \int_{(n-1)T}^{nT} x(t) dt \quad (2)$$

Approximating the definite integral by the ****trapezoidal rule**** (the area under the curve is approximated by the average height times the width T):

$$\int_{(n-1)T}^{nT} x(t) dt \approx \frac{T}{2} [x(nT) + x((n-1)T)] \quad (3)$$

Letting $y[n] = y(nT)$ and $x[n] = x(nT)$ yields the discrete difference equation:

$$y[n] - y[n-1] = \frac{T}{2} (x[n] + x[n-1]) \quad (4)$$

2.2 The Algebraic Mapping Operator

Taking the unilateral Z-transform of (4):

$$Y(z)(1 - z^{-1}) = \frac{T}{2} X(z)(1 + z^{-1}) \implies \frac{Y(z)}{X(z)} = \frac{T}{2} \frac{1 + z^{-1}}{1 - z^{-1}} \quad (5)$$

Comparing this discrete integrator transfer function with the continuous integrator $1/s$ in (1):

$$\frac{1}{s} \approx \frac{T}{2} \frac{1 + z^{-1}}{1 - z^{-1}} \quad (6)$$

Inverting both sides yields the fundamental **Bilinear Transformation**:

$$s = \frac{2}{T} \left(\frac{1 - z^{-1}}{1 + z^{-1}} \right) = \frac{2}{T} \left(\frac{z - 1}{z + 1} \right) \quad (7)$$

Solving for z gives the inverse mapping:

$$z = \frac{1 + \frac{T}{2}s}{1 - \frac{T}{2}s} \quad (8)$$

3. CONFORMAL MAPPING PROPERTIES & STABILITY

3.1 Mapping of the s-Plane to the z-Plane

Let $s = \sigma + j\Omega$ in (8):

$$z = \frac{1 + \frac{T}{2}(\sigma + j\Omega)}{1 - \frac{T}{2}(\sigma + j\Omega)} = \frac{(1 + \frac{\sigma T}{2}) + j\frac{\Omega T}{2}}{(1 - \frac{\sigma T}{2}) - j\frac{\Omega T}{2}} \quad (9)$$

Evaluating the magnitude squared $|z|^2$:

$$|z|^2 = \frac{(1 + \frac{\sigma T}{2})^2 + (\frac{\Omega T}{2})^2}{(1 - \frac{\sigma T}{2})^2 + (\frac{\Omega T}{2})^2} \quad (10)$$

We observe three decisive geometric properties:

1. **Left-Half Plane ($\sigma < 0$):** If $\sigma < 0$, then $(1 + \frac{\sigma T}{2})^2 < (1 - \frac{\sigma T}{2})^2$. Therefore, the numerator is strictly smaller than the denominator:

$$|z| < 1 \quad (\text{Strictly inside the unit circle}) \quad (11)$$

Stability Proof: Every stable analog pole in the left-half s -plane maps strictly inside the unit circle. The digital filter is **unconditionally stable**!

2. **Imaginary Axis ($\sigma = 0$):** If $\sigma = 0$, the numerator equals the denominator:

$$|z| = 1 \quad (\text{Exactly on the unit circle}) \quad (12)$$

The entire imaginary axis $s = j\Omega$ from $\Omega = -\infty$ to $\Omega = +\infty$ maps onto the unit circle $z = e^{j\omega}$ from $\omega = -\pi$ to $\omega = +\pi$.

3. **Right-Half Plane ($\sigma > 0$):** If $\sigma > 0$, $|z| > 1$, mapping outside the unit circle.

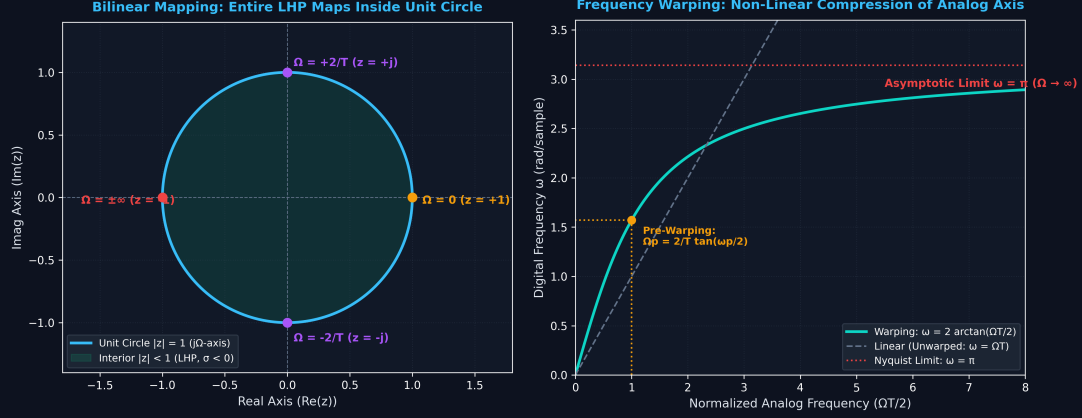


Figure 1: Bilinear Transformation characteristics. Left: One-to-one conformal mapping of the entire continuous s -plane into the discrete z -plane without spectral overlap. Right: The non-linear frequency warping curve $\omega = 2 \arctan(\Omega T/2)$, illustrating the compression of infinite analog frequencies into the finite digital interval $[0, \pi]$.

3.2 Total Elimination of Aliasing

Because the mapping between the continuous $j\Omega$ -axis and the discrete unit circle is **one-to-one and onto**:

- No point on the unit circle corresponds to more than one point on the analog frequency axis.
- **Zero Aliasing occurs!**
- Unlike Impulse Invariance, the Bilinear Transformation can be applied to design **any filter type**, including **Highpass, Bandpass, and Bandstop filters**!

4. FREQUENCY WARPING & THE PRE-WARPING PROTOCOL

4.1 Derivation of the Warping Equation

To find the relationship between digital frequency ω and analog frequency Ω , evaluate (7) on the unit circle ($z = e^{j\omega}$):

$$\begin{aligned} s = j\Omega &= \frac{2}{T} \left(\frac{1 - e^{-j\omega}}{1 + e^{-j\omega}} \right) = \frac{2}{T} \left(\frac{e^{-j\omega/2}(e^{j\omega/2} - e^{-j\omega/2})}{e^{-j\omega/2}(e^{j\omega/2} + e^{-j\omega/2})} \right) \\ &= \frac{2}{T} \left(\frac{2j \sin(\omega/2)}{2 \cos(\omega/2)} \right) = j \frac{2}{T} \tan\left(\frac{\omega}{2}\right) \end{aligned} \quad (13)$$

Equating imaginary parts yields the fundamental **Frequency Warping Formula**:

$$\Omega = \frac{2}{T} \tan\left(\frac{\omega}{2}\right) \quad (14)$$

Inverting (14) gives digital frequency as a function of analog frequency:

$$\omega = 2 \arctan\left(\frac{\Omega T}{2}\right) \quad (15)$$

Consequences of Frequency Warping:

- At low frequencies ($\omega \ll 1$), $\tan(\omega/2) \approx \omega/2 \implies \Omega \approx \omega/T$ (Linear relationship).
- As $\omega \rightarrow \pi$, $\tan(\omega/2) \rightarrow \infty \implies \Omega \rightarrow \infty$. The infinite analog frequency axis $[0, \infty)$ is non-linearly compressed into the finite digital interval $[0, \pi)$.
- This compression non-linearly distorts transition widths and frequency spacing.

4.2 The Principle of Frequency Pre-Warping

To compensate for frequency compression, the designer must ****pre-warp**** the desired digital critical frequencies (ω_p, ω_s) into analog critical frequencies (Ω_p, Ω_s) ****before** designing the analog prototype******:

$$\Omega_p = \frac{2}{T} \tan\left(\frac{\omega_p}{2}\right), \quad \Omega_s = \frac{2}{T} \tan\left(\frac{\omega_s}{2}\right) \quad (16)$$

When the analog prototype $H_a(s)$ is synthesized to meet Ω_p and Ω_s , the subsequent bilinear transformation maps these frequencies back to ω_p and ω_s ****with mathematical perfection****!

5 THE MATCHED Z-TRANSFORM (MZT) METHOD

5.1 Basic Concept

The **Matched Z-Transform (MZT)** is an alternative heuristic technique that maps the poles and finite zeros of an analog transfer function directly into discrete poles and zeros:

$$(s - p_k) \longrightarrow (1 - e^{p_k T} z^{-1}), \quad (s - z_k) \longrightarrow (1 - e^{z_k T} z^{-1}) \quad (17)$$

For an analog filter with M zeros and N poles ($M \leq N$):

$$H_a(s) = K \frac{\prod_{k=1}^M (s - z_k)}{\prod_{k=1}^N (s - p_k)} \implies H(z) = K_d \frac{\prod_{k=1}^M (1 - e^{z_k T} z^{-1})}{\prod_{k=1}^N (1 - e^{p_k T} z^{-1})} \quad (18)$$

5.2 Handling Zeros at Infinity

Most analog lowpass prototypes (such as Butterworth filters) have all-pole transfer functions where $M = 0$ and N zeros lie at $s = \infty$.

- Because $s = \infty$ corresponds to the digital Nyquist frequency $\omega = \pi$ ($z = -1$), the Matched Z-Transform inserts zeros at $z = -1$:

$$(1 + z^{-1})^{N-M} \quad (19)$$

- The gain constant K_d is then chosen to match the DC gain: $H(1) = H_a(0)$.
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6. COMPREHENSIVE COMPARISON OF IIR DESIGN METHODS

Table 1: Comprehensive Comparison of Analog-to-Digital IIR Transformation Techniques

Feature / Property	Impulse Invariance	Bilinear Transformation (BLT)	Matched Z-Transform (MZT)
Core Mapping Principle	$h[n] = T_s h_a(nT_s)$	$s = \frac{2}{T} \left(\frac{1-z^{-1}}{1+z^{-1}} \right)$	$(s-a) \rightarrow (1-e^{aT}z^{-1})$
Mathematical Basis	Time-domain sampling	Trapezoidal numerical integration	Direct pole-zero substitution
Stability Preservation	Guaranteed ($\text{Re}(s_k) < 0 \implies z_k < 1$)	Guaranteed ($\text{Re}(s_k) < 0 \implies z_k < 1$)	Guaranteed ($\text{Re}(s_k) < 0 \implies z_k < 1$)
Aliasing Occurrence	**Severe** (infinite spectral superposition)	**None** (one-to-one conformal mapping)	Mild aliasing present
Frequency Relationship	Linear: $\omega = \Omega T_s$	Non-linear: $\Omega = \frac{2}{T} \tan(\omega/2)$	Non-linear: $\omega = \Omega T_s$
Pre-Warping Needed?	No	**Yes** (mandatory for critical edges)	No
Permissible Filters	Band-limited LPF and BPF only	**Universal** : LPF, HPF, BPF, BSF**	LPF and BPF primarily
Mapping of Zeros	Zeros do **not** map algebraically	Zeros map via same algebraic rule	Zeros map directly to $z_k = e^{s_a k T}$

7. STEP-BY-STEP WORKED NUMERICAL EXAMPLES

7.1 Example 25.1: First-Order Analog to Digital BLT Transformation

Problem Statement: An analog lowpass filter is described by:

$$H_a(s) = \frac{2}{s+2} \quad (20)$$

Determine the digital filter transfer function $H(z)$ using the Bilinear Transformation with $T = 0.5$ seconds.

Solution:

Step 1: Formulate the BLT Substitution Operator With $T = 0.5$ s, $\frac{2}{T} = \frac{2}{0.5} = 4$. The substitution rule is:

$$s = 4 \left(\frac{1-z^{-1}}{1+z^{-1}} \right) \quad (21)$$

Step 2: Substitute s into $H_a(s)$

$$H(z) = \frac{2}{s+2} \Big|_{s=4\left(\frac{1-z^{-1}}{1+z^{-1}}\right)} = \frac{2}{4\left(\frac{1-z^{-1}}{1+z^{-1}}\right) + 2} \quad (22)$$

Multiplying numerator and denominator by $(1+z^{-1})$:

$$\begin{aligned} H(z) &= \frac{2(1+z^{-1})}{4(1-z^{-1}) + 2(1+z^{-1})} = \frac{2+2z^{-1}}{(4-4z^{-1}) + (2+2z^{-1})} \\ &= \frac{2+2z^{-1}}{6-2z^{-1}} \end{aligned} \quad (23)$$

Dividing numerator and denominator by 6:

$$\mathbf{H}(z) = \frac{\frac{1}{3} + \frac{1}{3}z^{-1}}{1 - \frac{1}{3}z^{-1}} = \frac{\mathbf{0.3333(1+z^{-1})}}{\mathbf{1-0.3333z^{-1}}} \quad (24)$$

Step 3: Verification of Pole Location and DC Gain

- Pole location: $z_1 = \frac{1}{3} \approx 0.3333 < 1$ ($\implies$ strictly stable!).
- Transmission Zero: At $z = -1$ ($\omega = \pi$), producing a transmission null at Nyquist!
- DC Gain ($z = 1$): $H(1) = \frac{0.3333(2)}{1-0.3333} = \frac{0.6667}{0.6667} = \mathbf{1.0} = H_a(0)$ (Exact DC gain match!).

7.2 Example 25.2: Complete Design of a Second-Order Digital Butterworth Filter

Problem Statement: Design a second-order digital Butterworth lowpass filter with a 3-dB cutoff frequency of $\omega_c = 0.2\pi$ rad/sample using the Bilinear Transformation with $T = 1$ second.

Solution:

Step 1: Pre-Warp the Digital Cutoff Frequency Using (16) with $T = 1$:

$$\Omega_c = \frac{2}{T} \tan\left(\frac{\omega_c}{2}\right) = 2 \tan\left(\frac{0.2\pi}{2}\right) = 2 \tan(0.1\pi) = 2 \tan(18^\circ) \quad (25)$$

Since $\tan(18^\circ) \approx 0.32492$:

$$\Omega_c = 2 \times 0.32492 = 0.6498 \text{ rad/s} \quad (26)$$

Step 2: Formulate the Normalized 2nd-Order Butterworth Analog Prototype The transfer function of a normalized 2nd-order Butterworth lowpass filter with cutoff Ω_c is:

$$H_a(s) = \frac{\Omega_c^2}{s^2 + \sqrt{2}\Omega_c s + \Omega_c^2} \quad (27)$$

Substituting $\Omega_c = 0.6498$:

- $\Omega_c^2 = (0.6498)^2 \approx \mathbf{0.4223}$
- $\sqrt{2}\Omega_c = \sqrt{2}(0.6498) \approx \mathbf{0.9190}$

Thus:

$$H_a(s) = \frac{0.4223}{s^2 + 0.9190s + 0.4223} \quad (28)$$

Step 3: Apply the Bilinear Transformation ($T = 1$) With $T = 1$, $s = 2 \left(\frac{1-z^{-1}}{1+z^{-1}} \right)$:

$$H(z) = \frac{0.4223}{\left[2 \left(\frac{1-z^{-1}}{1+z^{-1}} \right) \right]^2 + 0.9190 \left[2 \left(\frac{1-z^{-1}}{1+z^{-1}} \right) \right] + 0.4223} \quad (29)$$

Multiply numerator and denominator by $(1+z^{-1})^2$:

$$\text{Numerator} = 0.4223(1+z^{-1})^2 = 0.4223(1+2z^{-1}+z^{-2}) \quad (30)$$

$$\begin{aligned} \text{Denominator} &= 4(1-z^{-1})^2 + 1.8380(1-z^{-1})(1+z^{-1}) + 0.4223(1+z^{-1})^2 \\ &= 4(1-2z^{-1}+z^{-2}) + 1.8380(1-z^{-2}) + 0.4223(1+2z^{-1}+z^{-2}) \end{aligned} \quad (31)$$

Collecting like powers of z^{-1} :

- Constant term (z^0): $4 + 1.8380 + 0.4223 = \mathbf{6.2603}$
- Coefficient of z^{-1} : $-8 + 0 + 0.8446 = -\mathbf{7.1554}$
- Coefficient of z^{-2} : $4 - 1.8380 + 0.4223 = \mathbf{2.5843}$

Thus:

$$H(z) = \frac{0.4223(1+2z^{-1}+z^{-2})}{6.2603 - 7.1554z^{-1} + 2.5843z^{-2}} \quad (32)$$

Dividing numerator and denominator by 6.2603:

$$\mathbf{H(z)} = \frac{\mathbf{0.06746(1+2z^{-1}+z^{-2})}}{\mathbf{1-1.1430z^{-1}+0.4128z^{-2}}} \quad (33)$$

Verification:

- DC Gain ($z = 1$): $H(1) = \frac{0.06746(4)}{1-1.1430+0.4128} = \frac{0.26984}{0.2698} \approx \mathbf{1.0}$ (Exact 0 dB gain at DC!).
- Nyquist Gain ($z = -1$): $H(-1) = 0$ (Transmission zero at $\omega = \pi$).
- Stability: Roots of denominator have magnitude $|z| = \sqrt{0.4128} \approx 0.6425 < 1$ ($\implies$ strictly stable!).

7.3 Example 25.3: Design via Matched Z-Transform

Problem Statement: Apply the Matched Z-Transform method to convert:

$$H_a(s) = \frac{1}{(s+1)(s+2)} \quad (34)$$

into a digital filter with $T = 0.1$ s, adding zeros at $z = -1$ and matching DC gain.

Solution:

Step 1: Map Continuous Poles to Discrete Poles Analog poles: $s_1 = -1, s_2 = -2$. With $T = 0.1$ s:

$$z_1 = e^{s_1 T} = e^{-0.1} \approx 0.9048, \quad z_2 = e^{s_2 T} = e^{-0.2} \approx 0.8187 \quad (35)$$

Denominator polynomial:

$$D(z) = (1 - 0.9048z^{-1})(1 - 0.8187z^{-1}) = 1 - 1.7235z^{-1} + 0.7408z^{-2} \quad (36)$$

Step 2: Insert Zero at $z = -1$ Since $H_a(s)$ has $N = 2$ poles and $M = 0$ finite zeros, we add a zero at $z = -1$:

$$H(z) = K_d \frac{1 + z^{-1}}{(1 - 0.9048z^{-1})(1 - 0.8187z^{-1})} \quad (37)$$

Step 3: Normalize DC Gain Analog DC Gain: $H_a(0) = \frac{1}{(1)(2)} = 0.5$. Digital DC Gain at $z = 1$:

$$H(1) = K_d \frac{2}{(1 - 0.9048)(1 - 0.8187)} = K_d \frac{2}{(0.0952)(0.1813)} = K_d \frac{2}{0.01726} = 115.87K_d \quad (38)$$

Equating to $H_a(0) = 0.5$:

$$115.87K_d = 0.5 \implies K_d = \frac{0.5}{115.87} \approx \mathbf{0.004315} \quad (39)$$

Final transfer function:

$$\mathbf{H(z) = \frac{0.004315(1 + z^{-1})}{1 - 1.7235z^{-1} + 0.7408z^{-2}}} \quad (40)$$

8. FACULTY INSIGHTS, COMMON PITFALLS & EXAM TRAPS

- **Pitfall 1: Forgetting to Pre-Warp Cutoff Frequencies:** Students frequently design the analog prototype using the raw digital frequency $\Omega = \omega/T$. Because frequency warping compresses frequencies non-linearly, the resulting digital filter will have a cutoff frequency significantly lower than required. **Always pre-warp:** $\Omega_c = \frac{2}{T} \tan(\omega_c/2)$.
- **Pitfall 2: Algebraic Errors in $(1 + z^{-1})^2$ Multiplication:** When clearing fractions after substituting $s = \frac{2}{T} \left(\frac{1-z^{-1}}{1+z^{-1}} \right)$, students often expand $(1 - z^{-1})(1 + z^{-1})$ as $1 - 2z^{-1} + z^{-2}$ instead of $1 - z^{-2}$, causing complete corruption of the denominator coefficients.

- **Pitfall 3: Assuming BLT Preserves Analog Phase Linearity:** The Bilinear Transformation non-linearly warps the frequency axis. Therefore, even if an analog filter had approximately linear phase, **BLT distorts the phase response**. Linear phase can never be achieved in an IIR filter designed via BLT!
- **Pitfall 4: Confusion Over Sampling Period T :** In university exams, if T is not explicitly specified, students often assume they cannot solve the problem. In reality, $T = 1$ or $T = 2$ can be chosen arbitrarily because T cancels out completely in the normalized equations when pre-warping is performed consistently!

9 9. QUICK REVISION & SUMMARY CHEAT-SHEET

Table 2: Bilinear Transformation and Matched Z-Transform Quick Reference Sheet

Concept / Parameter	Mathematical Definition / Formula	Key Pedagogical Note
BLT Substitution	$s = \frac{2}{T} \left(\frac{1-z^{-1}}{1+z^{-1}} \right)$	Replaces s directly in $H_a(s)$
Inverse BLT Mapping	$z = \frac{1+sT/2}{1-sT/2}$	Maps continuous s to discrete z
Frequency Warping	$\Omega = \frac{2}{T} \tan(\omega/2)$	Non-linear mapping between Ω and ω
Pre-Warping Formula	$\Omega_c = \frac{2}{T} \tan(\omega_c/2)$	Mandatory step before analog prototype design
Aliasing Behavior	**Zero Aliasing**	One-to-one mapping of $j\Omega$ onto unit circle
Filter Suitability	Universal: LPF, HPF, BPF, BSF	Valid for all standard filter configurations
MZT Pole Rule	$(s - p_k) \longrightarrow (1 - e^{p_k T} z^{-1})$	Direct exponential mapping of poles
MZT Zero Rule	Add $(1 + z^{-1})$ for zeros at $s = \infty$	Creates transmission null at Nyquist